

# Sylvia Grzegorzczuk

Full Stack Engineer & AI Architect | LLM Engineering | PHP, React, C#, SQL | System Design

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Open to B2B | Remote-first | Flexible for Global Timezones

Results-driven Full Stack Developer with over 5 years of experience in architecting scalable enterprise applications using PHP, C#, and React. Expert in integrating AI-driven workflows and LLM orchestration into production environments. Specialized in designing cost-optimized AI pipelines, performance tuning, and building resilient system architectures.

## CORE COMPETENCES

- **AI & LLM Engineering:** LLM-as-a-Judge Pattern, RAG Architectures (pgvector, Pinecone, Chroma DB), Prompt Engineering, Multimodal AI Pipelines, LangChain, LLM Evaluation (Ragas), Token-Usage Optimization.
- **Backend Development:** FastAPI, Python, PHP (Laravel, Symfony, Slim, Core), C# (.NET), Node.js, SQL (PostgreSQL, PL/pgSQL, PL/SQL), pgvector, Firebase, Deterministic Data Modeling.
- **Data & Databases:** PostgreSQL (pgvector), SQL Server, PL/SQL, Deterministic Data Modeling, Firebase.
- **Frontend Development:** React.js, Vue 3, TypeScript, Next.js, JavaScript (ES6+), HTML5, CSS3/SCSS/SASS, Pixi.js, GSAP, Material UI, Web Workers.
- **DevOps & Security:** OAuth 2.0, JWT, 2FA, bcrypt, Docker, Git, Bitbucket, Webpack, GCP (Google Cloud Platform).
- **Testing & Tools:** Jest, React Testing Library, Performance Tuning, Jira, Figma, CMS, PSPP (Data Analysis).
- **Methodologies:** Agile, Scrum, Clean Code Architecture, System Design, Technical Documentation, AI-assisted development (Copilot/Cursor), Exponential Backoff Strategies.
- **Soft Skills:** Strong communication & client collaboration, problem-solving, adaptability, intercultural teamwork, time management.

## WORK EXPERIENCE

### Full Stack Engineer • Self-employed

10/2024 – present

- **AI & Architecture:** Architected multi-stage AI pipelines using the LLM-as-a-Judge pattern for automated multimodal analysis.
- **RAG Architecture:** Engineered a decoupled, context-aware RAG platform utilizing FastAPI and Vue 3, enabling automated, hallucination-free retrieval from complex domain-specific medical PDFs.
- **Vector Pipeline & MLOps:** Built a deterministic data ingestion pipeline using LangChain and Chroma DB with gemini-embedding-001; implemented an offline evaluation suite using Ragas to mathematically benchmark Faithfulness and Answer Relevance before staging.
- **Secure API Design & Deployment:** Designed a cost-optimized authentication and token pattern using client-side localStorage and dynamic injection via custom X-Gemini-API-Key request headers; deployed the microservices architecture seamlessly onto Render.com and Netlify.
- **Web Development & Design:** Developed high-performance, SEO-optimized web applications and digital products, focusing on clean architecture, client collaboration and modern user experiences.
- **Design Systems:** Built and maintained scalable Design Systems for clients to ensure UI/UX consistency across platforms and accelerate frontend development cycles.
- **SEO Audits & Optimization:** Conducted technical SEO audits and implemented performance-tuning strategies, **resulting in a 40% increase in organic traffic** and improving Core Web Vitals (LCP) **from 'Poor' to 'Good'** within 3 months for key client projects.
- **Operational Efficiency:** Implemented hybrid validation models to optimize OpEx and integrated exponential backoff strategies (Tenacity) for API resilience.
- **Advanced Frontend:** Optimized performance via client-side image compression using Web Workers, enhancing TTFT and reducing payload sizes.

### Full Stack Engineer • Softarchitect • Warsaw, Poland

08/2021 – 09/2024

- **Enterprise System Architecture:** Independently designed and integrated large-scale, scalable features into high-traffic enterprise environments. Focused on maintaining system reliability and seamless operation while adapting new modules to complex legacy infrastructures.

- **Legacy Code Refactoring & Debugging:** Expertly debugged, prototyped, and refactored existing codebases to resolve deep-seated software issues and improve overall system maintainability. Significantly enhanced application performance by optimizing inefficient logic and improving database query execution.
- **Full Stack Development & Data:** Developed and maintained robust front-end and back-end components using PHP, C#, JavaScript (ES6+), and SQL. Engineered efficient data transformation pipelines to process and analyze large datasets, improving application logic and backend efficiency.
- **Security & API Orchestration:** Designed and implemented secure enterprise authentication solutions, including OAuth 2.0, JWT, and Two-Factor Authentication (2FA). Developed high-performance RESTful APIs to ensure seamless data communication between distributed client-server architectures.
- **Mentorship & Quality Assurance:** Led code reviews and provided technical mentorship to cross-functional teams, ensuring adherence to Clean Code architecture and industry best practices. Conducted comprehensive system testing to deliver high-quality, stable software releases.
- **Stakeholder Management:** Maintained active collaboration with clients and stakeholders to gather technical requirements, provide project updates, and ensure the delivery of optimized, user-centric solutions.

## EDUCATION

- **Bachelor's Degree in Human Nutrition and Dietetics** - Medical University of Lublin – 2017-2020
- Completed **7 international training** courses focused on developing core competencies in youth engagement, educational techniques, project management, and intercultural communication.

## LANGUAGES

- **Polish** – native
- **English** – advanced
- **Spanish** – intermediate, currently progressing through active language learning

I agree to the processing of personal data provided in this document for realising the recruitment process pursuant to the Personal Data Protection Act of 10 May 2018 (Journal of Laws 2018, item 1000) and in agreement with Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation).